

1(a)(i)	species <u>of tree</u> or type <u>of tree</u> ;	1
1(a)(ii)	any two from: 1 (same) number of / 3, plantations <u>for each tree species</u> ; 2 (same) size/area of, plot/(sampling)area / 36 m x 36 m / 1296m² ; 3 (same / used a) belt transect (same method in all plots/plantations) ; 4 (same) months<u> / seasons</u> (of the year) or sampling at the same times<u> ;</u>	2

Question	Answer	Marks
1(b)	any five from: <i>using a belt transect:</i> 1 use of tape / line / string, <u>and</u> quadrat (to create belt transect) ; 2 selecting a <u>start point</u> for the, transect / placement of tape / line / string or all transects same length, in plots / plantations / repeats ; 3 same (size) quadrat ; 4 sample at / use quadrat at, regular intervals / stated distances along the, tape / line / string / transect or continuous belt transect ; <i>collecting data:</i> 5 method to identify each of the plant (species in the quadrat) ; 6 count / record / note the <u>number of</u> (individuals), <u>each</u> plant species (n) (in each quadrat / at each sampling point) or estimate percentage (%) cover of <u>each</u> plant species (in each quadrat) ; 7 use at least 3 <u>different</u> transects (in each plot) ; 8 named hazard <u>and</u> risk <u>and</u> precaution ; e.g. (see table below)	5

Question	Answer			Marks
1(b)	hazard	risk	precaution	
	plants / fungi	falling trees / branches	hard hat	
		thorns	gloves / goggles / mask / long trousers / PPE	
		irritant / allergy infection (spores)	antihistamines / cover skin / first aid equipment	
	(named) animal(s) parasite	bites / infection / attack / stings / allergy	antivenom / antihistamine / trained, professionals / medics, available / insect repellent / gloves / goggles / mask / PPE / cover skin	
	plantation terrain	trip hazard	correct footwear / watch where you are walking / use walking sticks	
		getting lost	work in group / maps / use GPS / guide	
	poachers	being attacked / shot	trained, professionals/ medics, available	
1(c)	any one from: (does not support) 1. acacia and mango have the <u>same means</u> or acacia and mango have <u>similar, <i>D</i> values / index</u> or mango and invasive species have overlapping, <i>D</i> values / index / means +/- SD ; (does support / partially support) 2. <u>sal tree</u> has the <u>highest, <i>D</i> value / index / mean</u> or sal tree has greater, <i>D</i> values / Index / mean, than both alien species ;			1

Question	Answer			Marks
1(d)(i)	concentration / mg cm ⁻³	volume of stock solution / cm ³	volume of (distilled) water / cm ³	2
	7.5	37.5	12.5 ;	
	2.5	12.5	37.5 ;	
1(d)(ii)	1 correct processes <u>and</u> correct figures ; 2 (correct) answer to 3 significant figures ; ecf if mp1 is incorrect 3 decrease / minus (number) ;			3

Question	Answer	Marks																											
1(e)	any three from: <table border="1"> <thead> <tr> <th></th><th>improvement</th><th>impact</th></tr> </thead> <tbody> <tr> <td>1</td><td>More, intermediate / smaller values / intervals appropriate examples accepted</td><td>better idea of the, trend / pattern / effect ;</td></tr> <tr> <td>2</td><td>exclude anomalous results</td><td>improve reliability ;</td></tr> <tr> <td>3</td><td>repeat reading for 2.5 / 5.0 / 7.5 mg cm⁻³ (<i>any of these 3 might be anomalous</i>)</td><td>check the anomaly / improve reliability ;</td></tr> <tr> <td>4</td><td>repeat investigation <u>at least twice</u> or repeat with more marigolds</td><td>better idea of trend / improve reliability / reduce effect of anomalies ;</td></tr> <tr> <td>5</td><td>(calculate) standard error / standard deviation / confidence intervals / statistical analysis</td><td>check for significance (<i>difference between the means</i>) / checking accuracy ;</td></tr> <tr> <td>6</td><td>use marigold <u>clones</u></td><td>exclude genetic effects / improve validity ;</td></tr> <tr> <td>7</td><td>use Vernier calliper</td><td>more precise measurements ;</td></tr> <tr> <td>8</td><td>measure the mass of all the roots</td><td>improve reliability ;</td></tr> </tbody> </table>		improvement	impact	1	More, intermediate / smaller values / intervals appropriate examples accepted	better idea of the, trend / pattern / effect ;	2	exclude anomalous results	improve reliability ;	3	repeat reading for 2.5 / 5.0 / 7.5 mg cm ⁻³ (<i>any of these 3 might be anomalous</i>)	check the anomaly / improve reliability ;	4	repeat investigation <u>at least twice</u> or repeat with more marigolds	better idea of trend / improve reliability / reduce effect of anomalies ;	5	(calculate) standard error / standard deviation / confidence intervals / statistical analysis	check for significance (<i>difference between the means</i>) / checking accuracy ;	6	use marigold <u>clones</u>	exclude genetic effects / improve validity ;	7	use Vernier calliper	more precise measurements ;	8	measure the mass of all the roots	improve reliability ;	3
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1(f)	any two from: (same) mass of soil ; - sample from same depth ; - sampld at the same time ; - sample from same, distance from roots / position in the pot ;	2																											